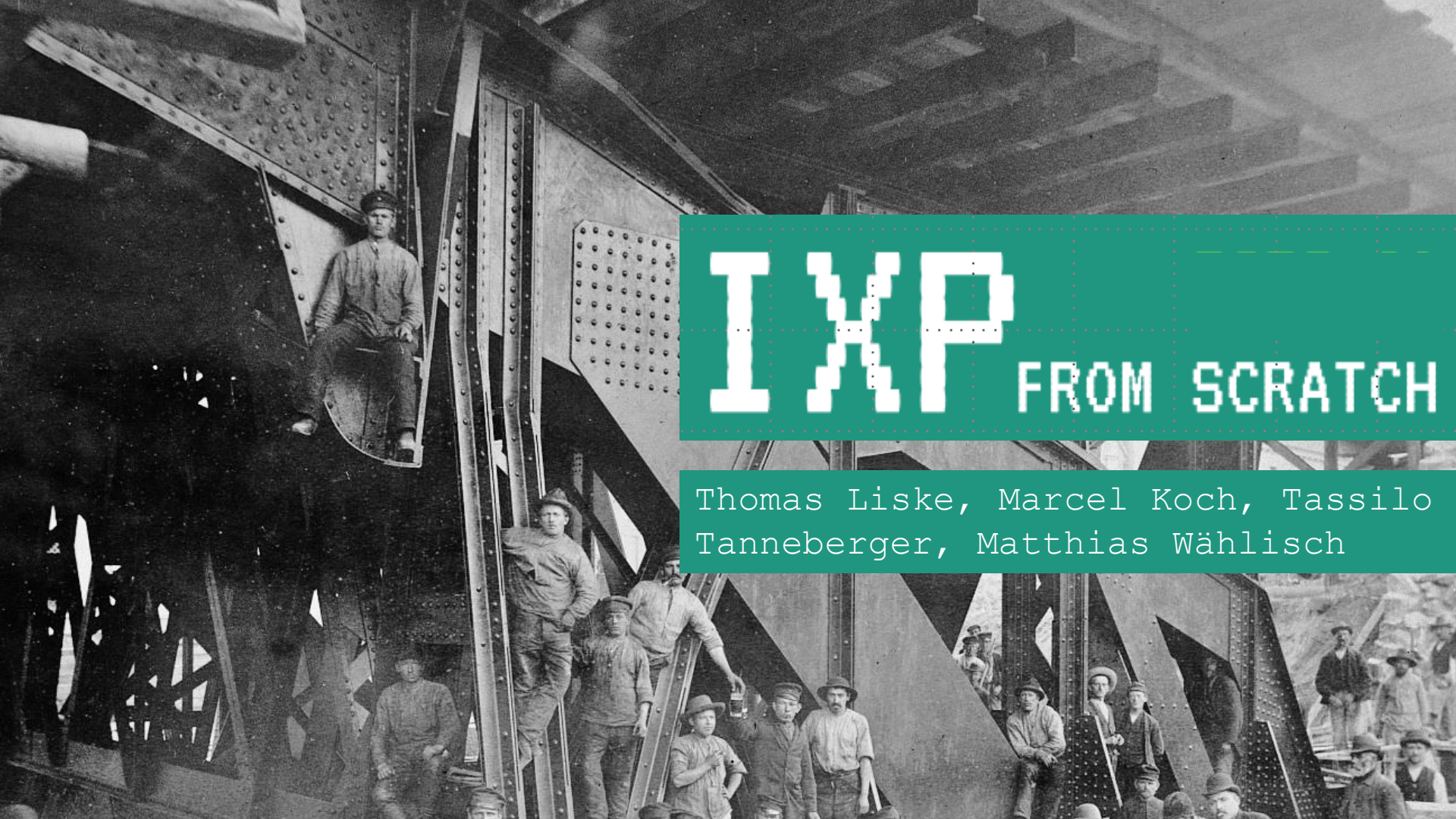




IXP FROM SCRATCH

Thomas Liske, Marcel Koch, Tassilo
Tanneberger, Matthias Wählisch

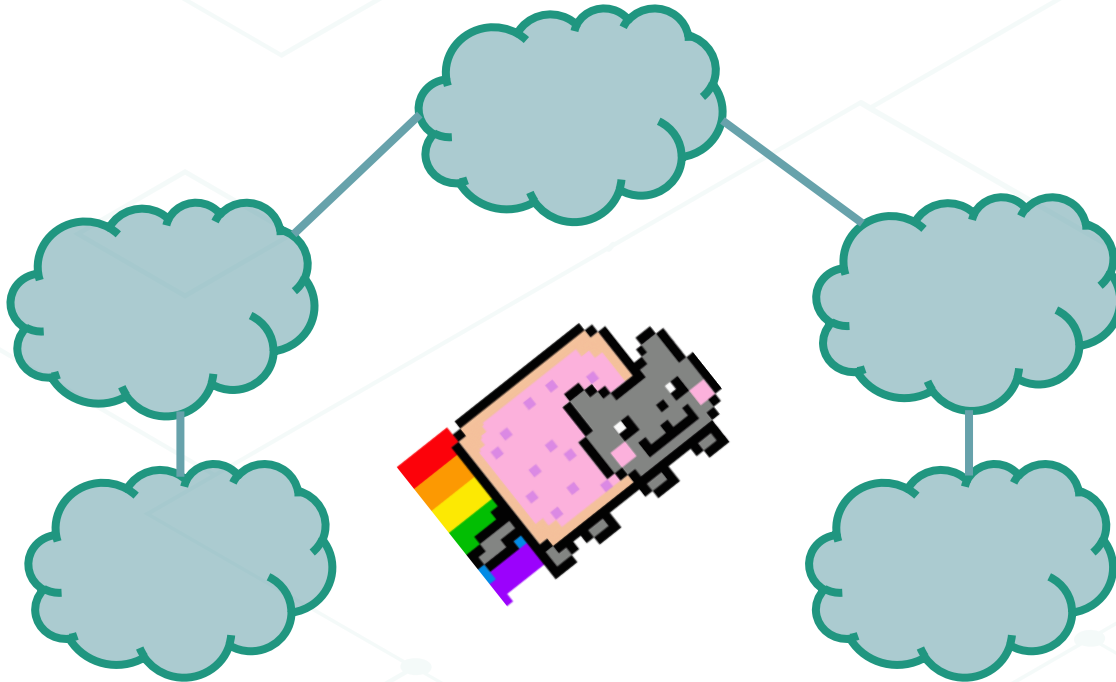
What is an Internet eXchange?



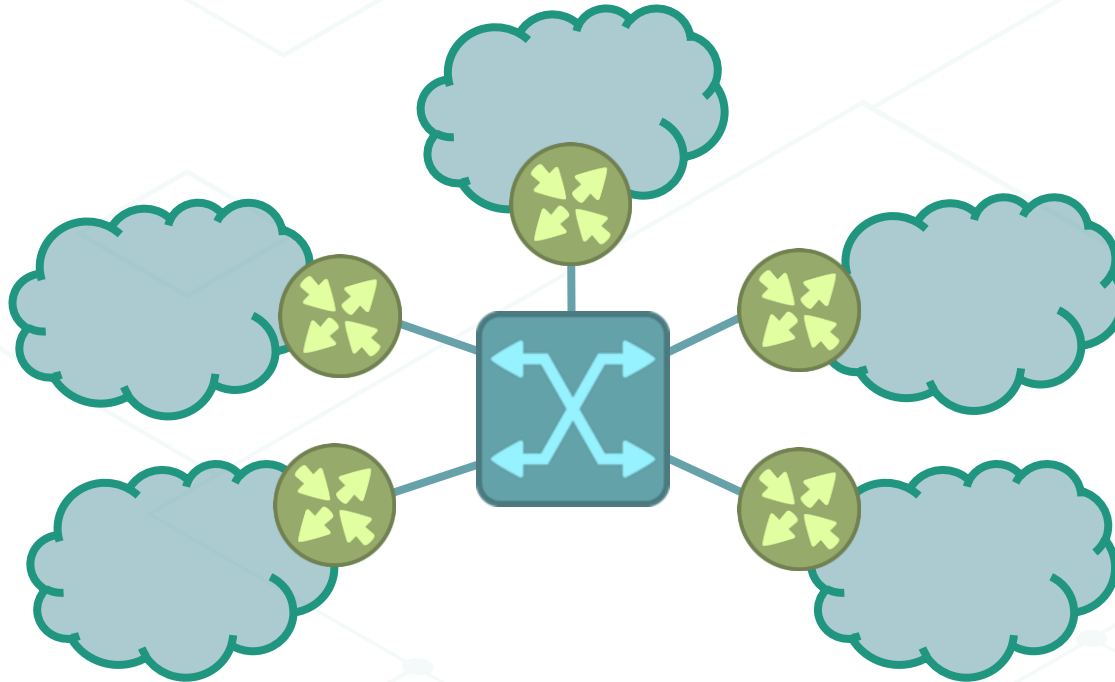
What is an Internet eXchange?



What is an Internet eXchange?



What is an Internet eXchange?



Interconnects Internet stakeholders locally by providing a switching infrastructure.

What is an Internet eXchange?



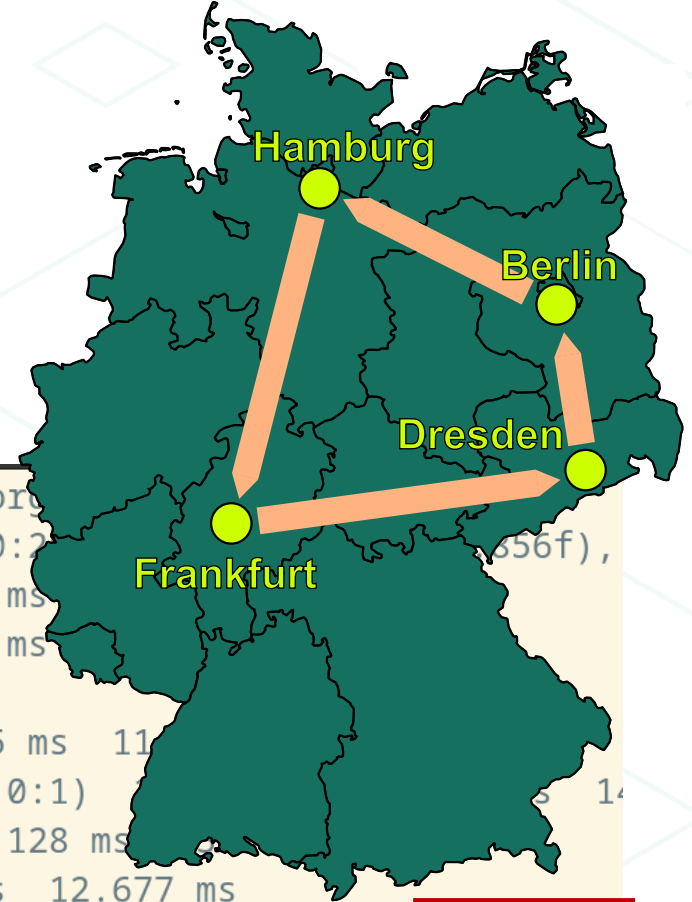
Interconnects Internet stakeholders **locally** + community building.

An IX is more than peering!

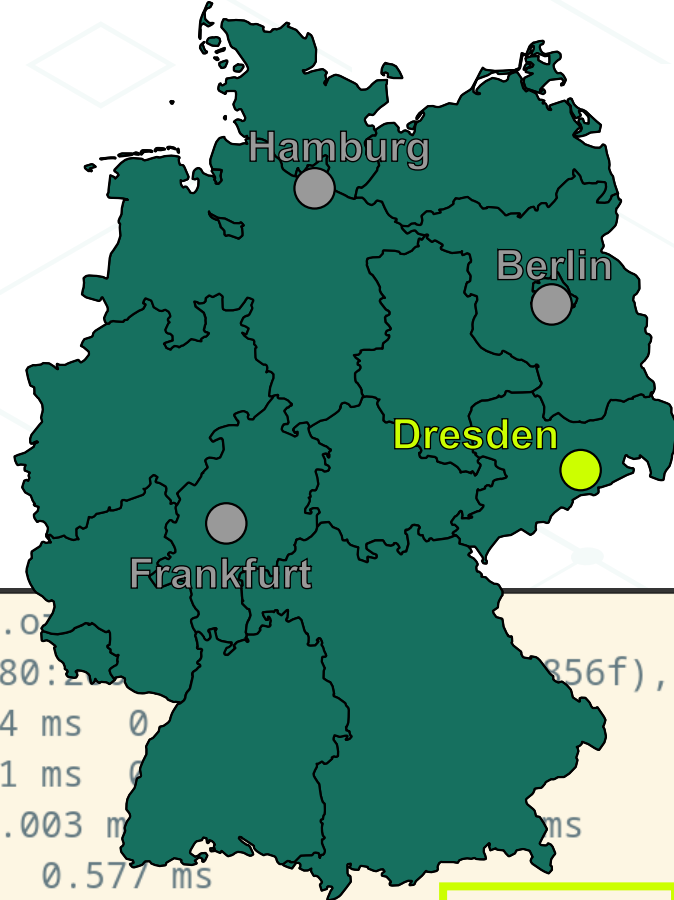
Why does a city, e.g., need an IX? Before DD-IX.

```
[root@micky:~]# traceroute upstream4.core.zentralwerk.org
traceroute to upstream4.core.zentralwerk.org (2a00:8180:2000:37:9e8:1602:c2f3:856f),
 1  dc-c1-cs1-v10.ibh.net (2a01:7700:0:1001::2)  1.679 ms
 2  rtr-c1-bg1.ibh.net (2a01:7700::3)  0.249 ms  0.251 ms
 3  * * *
 4  e0-35.core2.ham1.he.net (2001:470:0:134::2)  11.025 ms  11.025 ms
 5  ipv6.de-cix.fra.de.as16205.dsi.net (2001:7f8::3f4d:0:1)  13.141 ms  13.128 ms
 6  2a00:8180:0:1::4 (2a00:8180:0:1::4)  12.785 ms  12.807 ms  12.677 ms
 7  2a00:8180::18 (2a00:8180::18)  12.785 ms  12.807 ms  12.677 ms
 8  upstream4.core.zentralwerk.org (2a00:8180:2000:37:9e8:1602:c2f3:856f)  12.943 ms
```

[root@micky:~]#



Why does a city, e.g., need an IX? After DD-IX.



```
[root@micky:~]# traceroute upstream4.core.zentralwerk.org
traceroute to upstream4.core.zentralwerk.org (2a00:8180:2000:37:9e8:1602:c2f3:856f),
 1  dc-c1-cs1-v10.ibh.net (2a01:7700:0:1001::2)  0.744 ms  0
 2  rtr-c1-bg1.ibh.net (2a01:7700::3)  0.235 ms  0.231 ms
 3  dsinet.cc.peerdd-ix.net (2001:7f8:79::3f4d:1)  1.003 ms
 4  2a00:8180::18 (2a00:8180::18)  0.623 ms  0.604 ms  0.577 ms
 5  upstream4.core.zentralwerk.org (2a00:8180:2000:37:9e8:1602:c2f3:856f)  0.757 ms
[root@micky:~]#
```


Dresden Internet Exchange

2005 past efforts: IX in Dresden

2023 DD-IX e.V. gets formed



Dresden Internet Exchange

- 2024 operational
- 2024 multiple
"Open Tech Meetups"
- 2024 3 peers
- 2024 first Gbits/s peek
- 2024 More peers wait for X-Connect
- 2024 Official opening in Nov. 2024



IXPs are often community projects





Keep Local Traffic Local

IXP from Scratch

Published

1. Building a New IX
2. Network and Security Design
3. The Peering LAN

In progress

- Automation
- Monitoring & Traffic Analysis
- <your input>

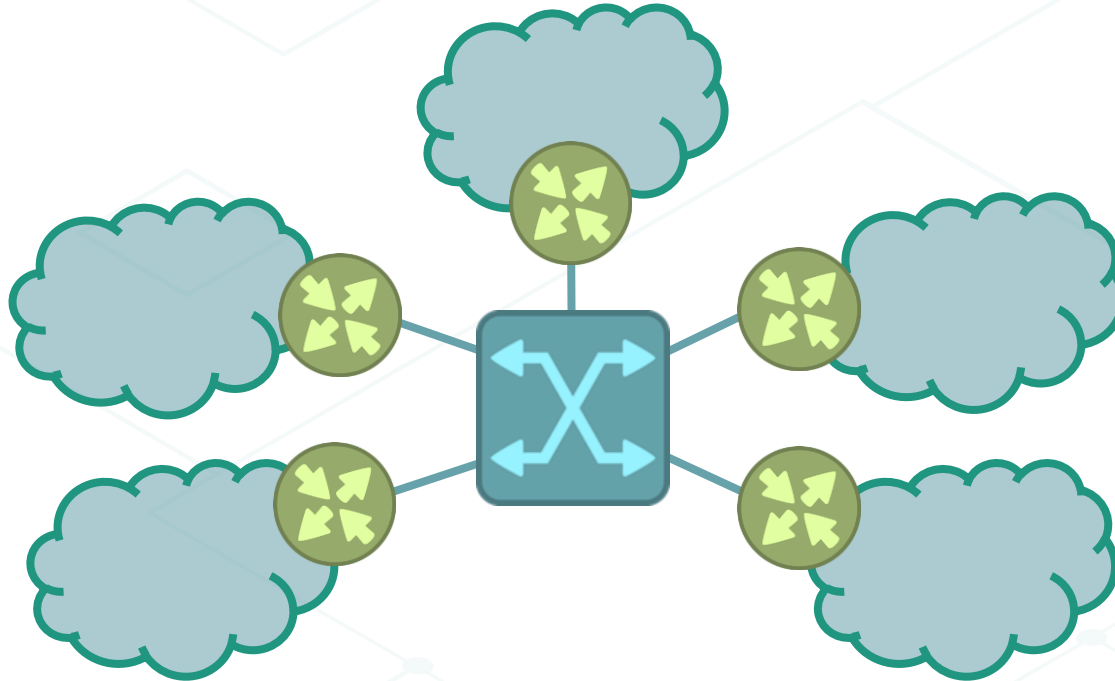
First RIPE Labs article:

<https://labs.ripe.net/search/?tags=ixp-from-scratch>



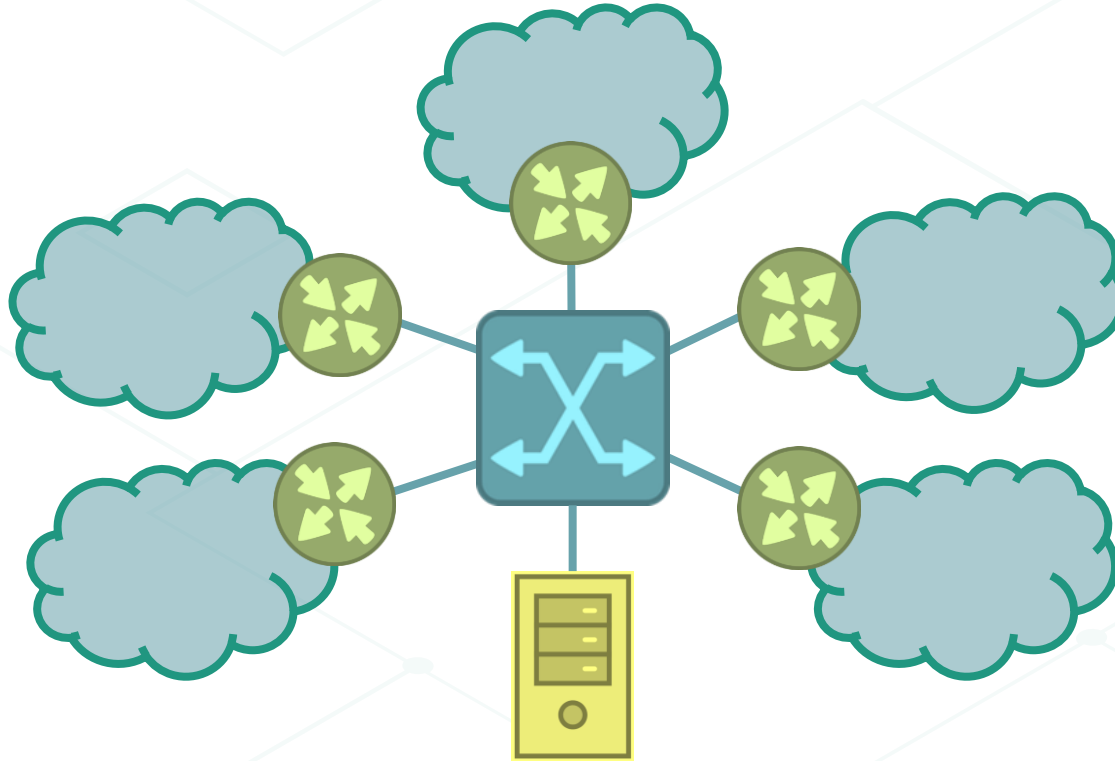
Which services does an IXP provide?

Which core services does an IXP provide?



An IXP provides layer 2 based on multiple switches.

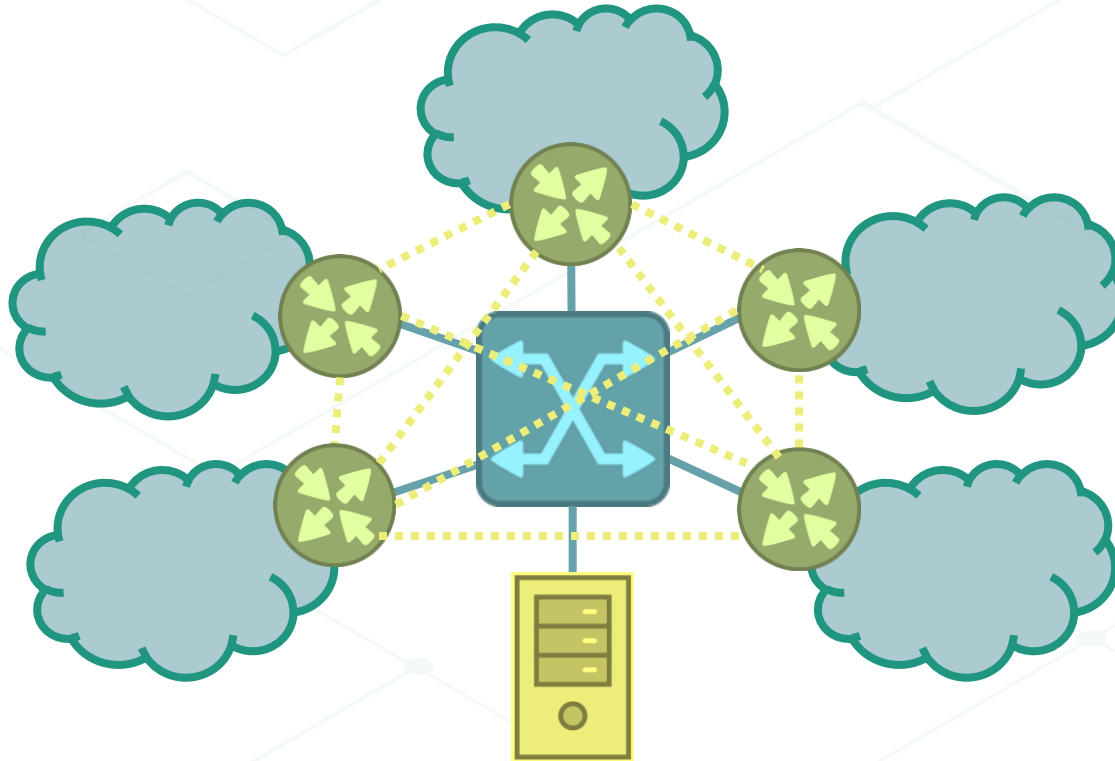
Which core services does an IXP provide?



An IXP provides layer 2 based on multiple switches.

And a route server to ease peering.

Which core services does an IXP provide?

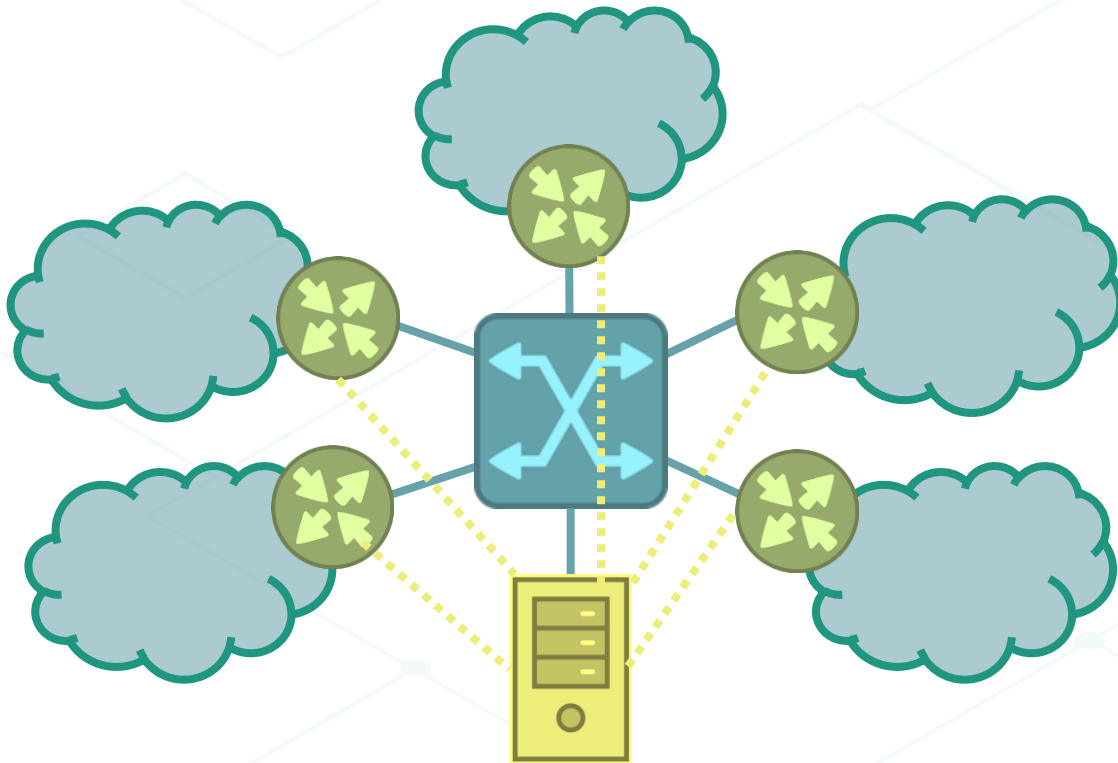


An IXP provides layer 2 based on multiple switches.

And a route server to ease peering.

Instead of full mesh ...

Which core services does an IXP provide?



An IXP provides layer 2 based on multiple switches.

And a route server to ease peering.

Instead of full mesh ... use a single BGP session.

Which core services does an IXP provide?

VS.

```
router bgp 51224
  neighbor 2001:db8:19:1::3ff6:10 remote-as 17380
  neighbor 2001:db8:19:1::3ff6:11 remote-as 18341
  neighbor 2001:db8:19:1::3ff6:12 remote-as 12312
  neighbor 2001:db8:19:1::3ff6:13 remote-as 14323
  neighbor 2001:db8:19:1::3ff6:14 remote-as 11334
  neighbor 2001:db8:19:1::3ff6:15 remote-as 12345
}
```

```
router bgp 51224
  neighbor 2001:db8:19:1::3ff6:1 remote-as 16374
}
```

Services of IXP Operations

- IXP services
 - peering LAN
 - IXP management
 - monitoring
- office and association
 - paperwork
 - accounting

Open-Source Software at DD-IX



To manage IXP services

IXP Manager - that was to be expected, wasn't it?

Alice-LG - our friendly looking glass for the route servers

To manage our network

Netbox - as IPAM and DCIM solution, not used for automation

Grafana & Prometheus - for observability and alerting

To manage our daily office work

Nextcloud - with OnlyOffice and draw.io integration

Vaultwarden - our password manager

Mailman3 - for community mailing lists (e.g. DDNOG)

Open-Source Software at DD-IX

When solutions are not available, we also found open-source software projects



IXpect

Auditing software for IXPs

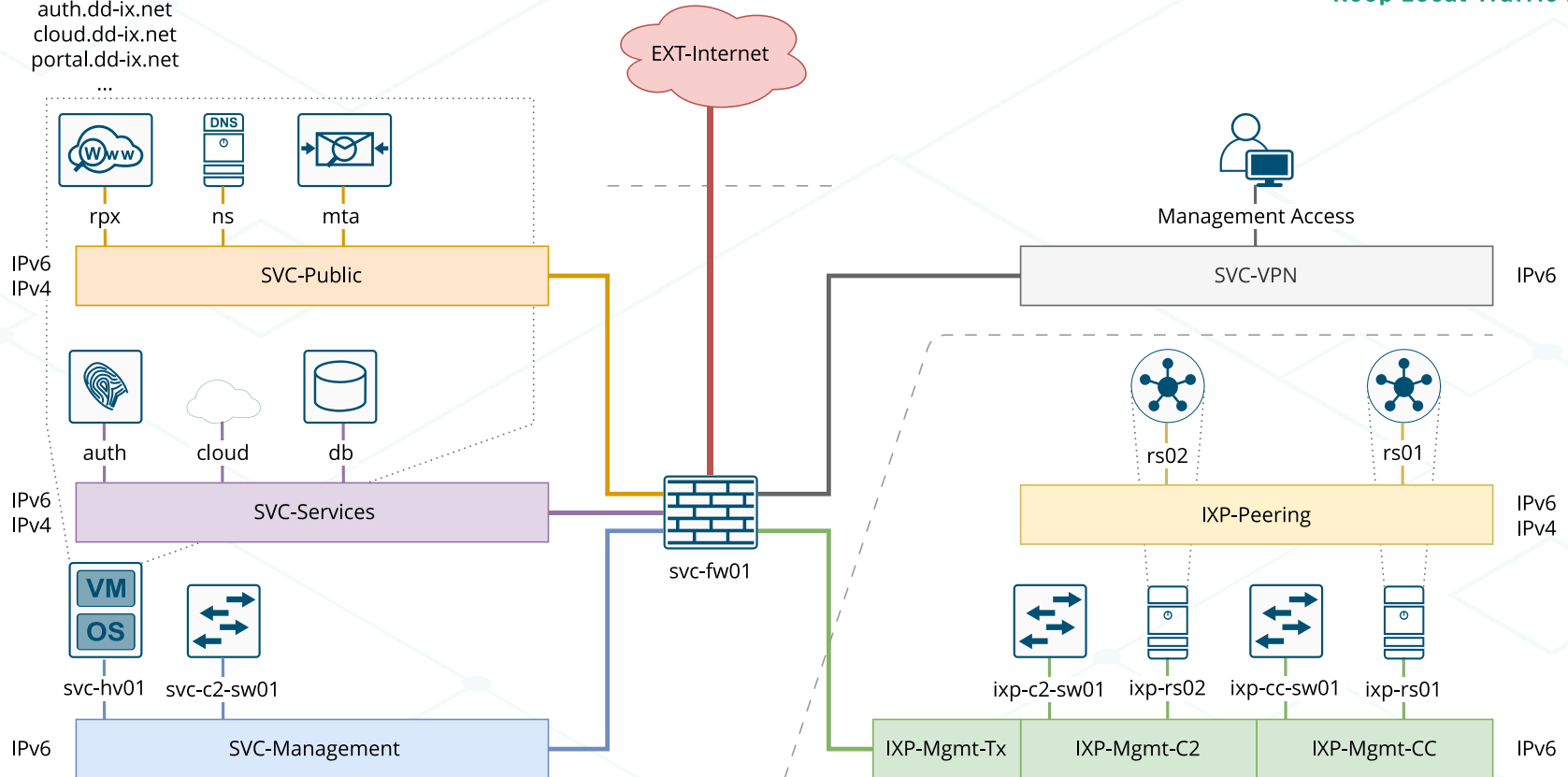
Checks whether L2/L3 traffic
conforms to IXP peering-policies

Available on <https://ixpect.net/>

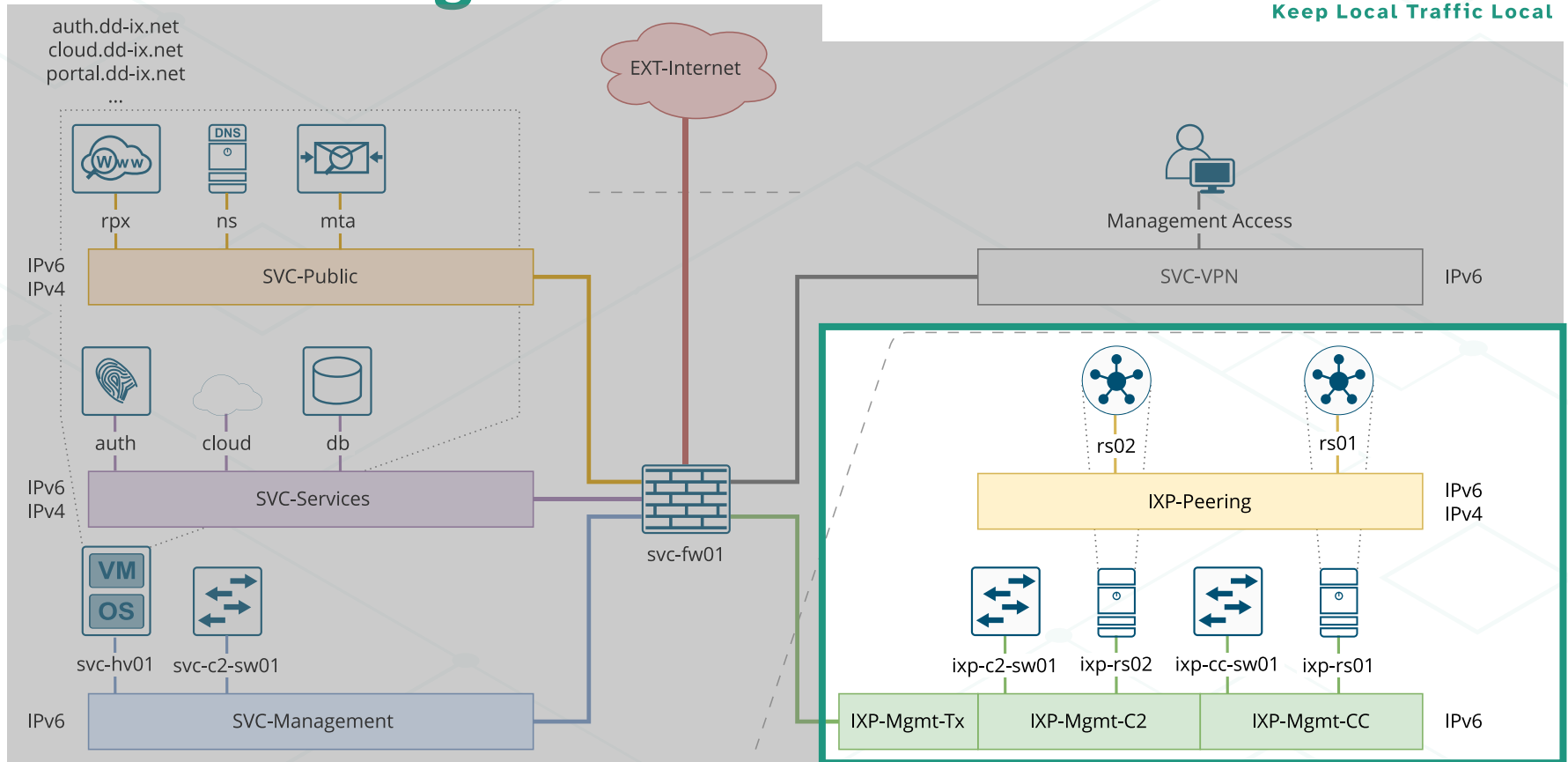
Network and Security Design

Network Segmentation

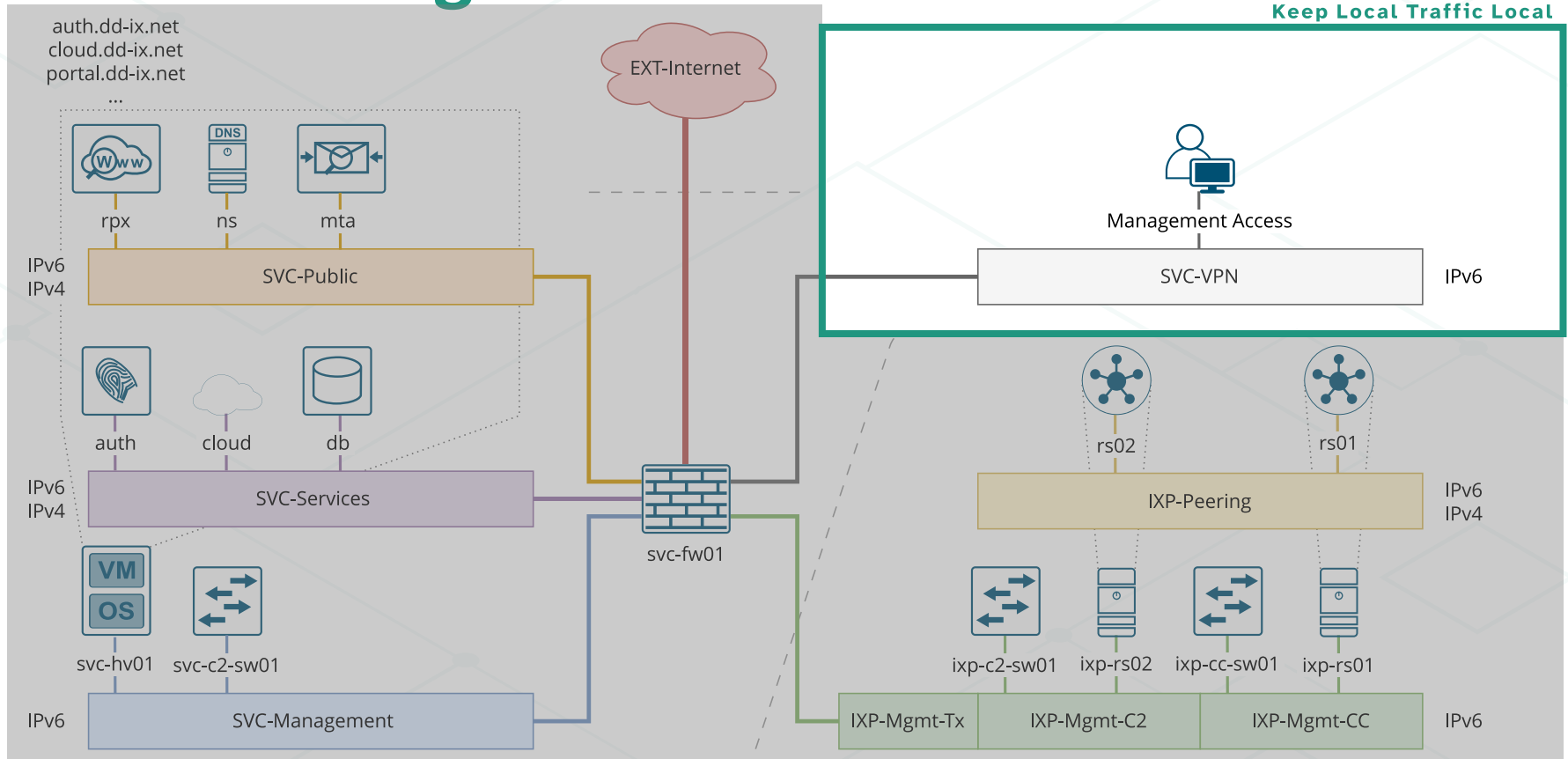
auth.dd-ix.net
cloud.dd-ix.net
portal.dd-ix.net
...



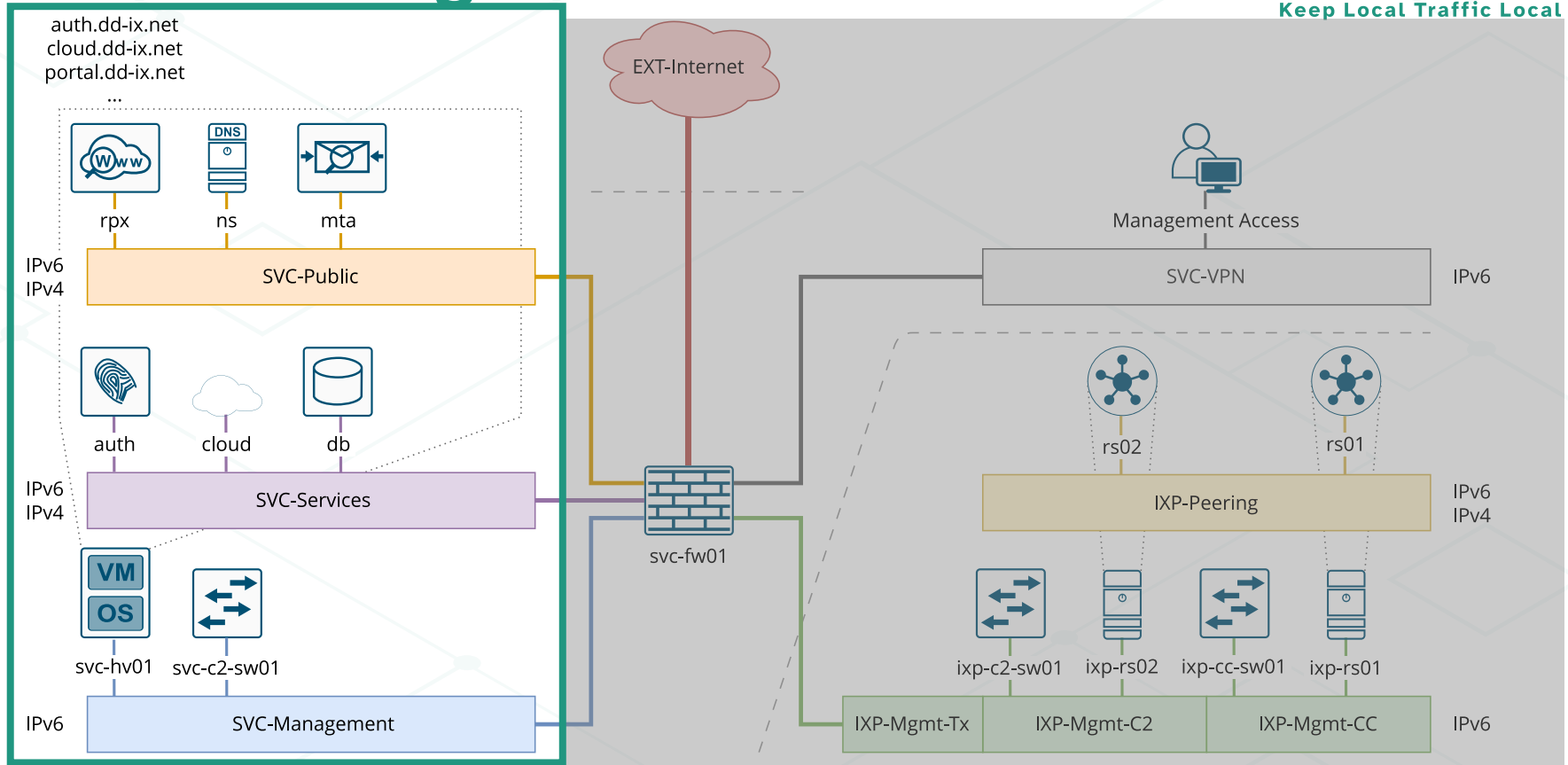
Network Segmentation



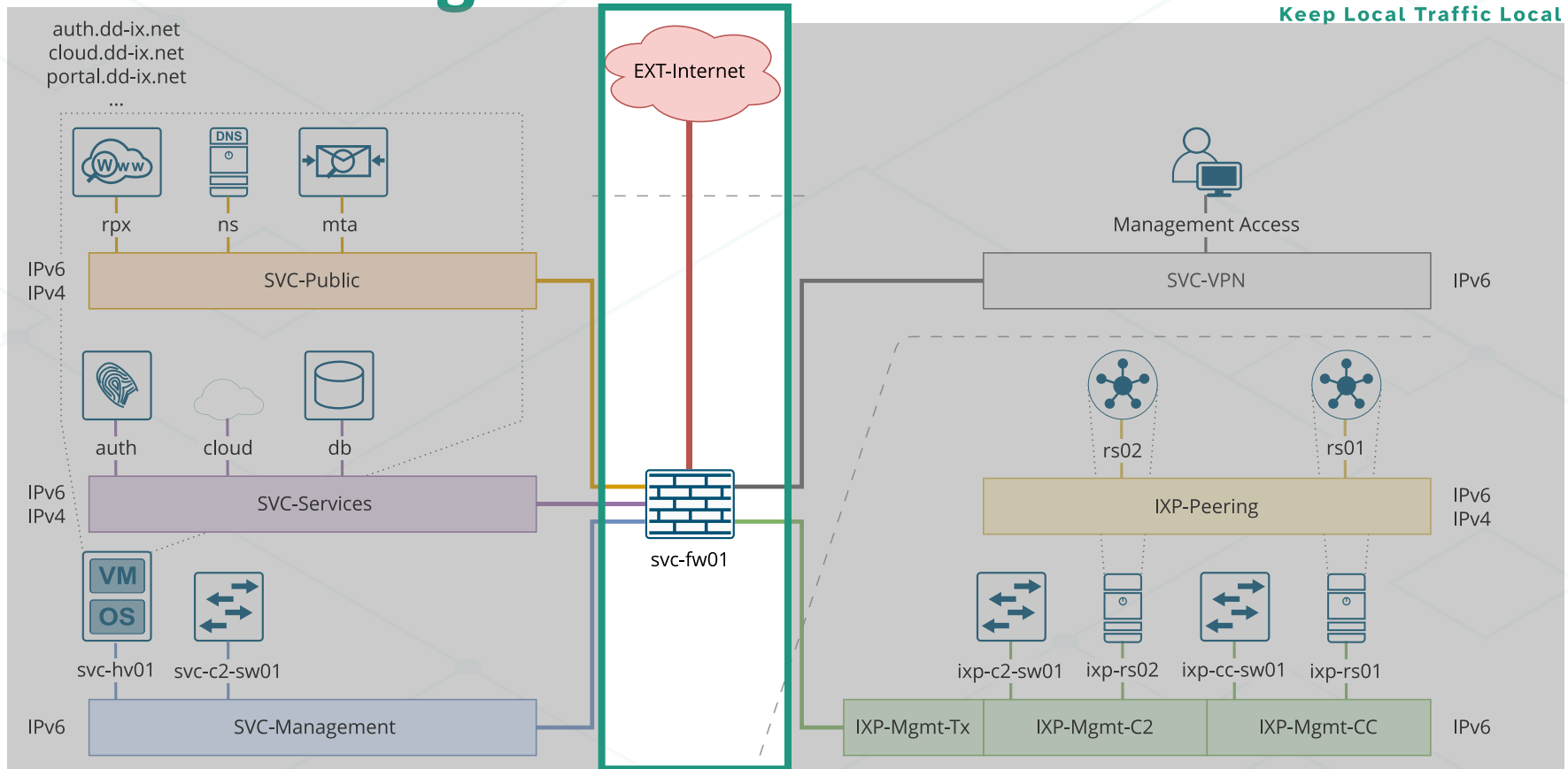
Network Segmentation



Network Segmentation



Network Segmentation



Addressing scheme

Use IPv6 addressing for all internal and external services

☐ Prefix	Utilization	Status	Description
☐ 2001:db8:dff0::/48	31.3%	Container	IBH PA
☐ • 2001:db8:dff0:3000::/52	0.0%	Container	IXP Public Services
☐ • 2001:db8:dff0:4000::/52	0.1%	Container	IXP Internal Services
☐ • 2001:db8:dff0:5000::/52	0.0%	Container	IXP EVPN Underlay
☐ • 2001:db8:dff0:6000::/52	0.0%	Container	SVC Public Services
☐ • 2001:db8:dff0:7000::/52	6.3%	Container	SVC Internal Services

The advantage: we can recognize the zone from the network ID of an IPv6 address without consulting our IP address management. It is, therefore, much easier for people to work with IPv6 addresses than with IPv4 addresses 😎.

IPv6-only, we've tried very hard

- ☹️ some external APIs from IXP Manager and arouteserver are IPv4-only
- ☹️ NixOS heavily relies on GitHub, GitHub is missing AAAA records...

=> We still need to use some private v4 addresses (RFC1918) ☹️

```
> host -t aaaa github.com  
github.com has no AAAA record
```

Firewall and routing

Design of access control lists

- **Firewall is based on nftables**
 - Allows to write dual stack access rules
- **Splitting ACLs into sub-ACLs based on security zone**
 - Split the access rules into sub-ACLs for each tuple of source and destination zone.
 - A sub-ACL always enforces a final decision: they all should have a final `deny any any` rule.
 - The inbound and outbound interfaces allow to map the corresponding source and destination zones.
 - From the main ACL, the corresponding sub-ACL is only called based on the zone tuple.
 - The sub-ACLs are named and ordered by the source and destination zone in the ruleset file for reasons of clarity.
- **Divide and conquer principle makes it easy to maintain even large firewall policies**
 - To add or find a rule, we only need to know the source and destination zones to locate the corresponding sub-ACL, which is usually very easy to understand.
 - Additional advantage: much lower risk of writing rules that allow for more than intended.

```

637 chain zone_services_adm {
638     # grafana: allow prometheus access
639     ip6 saddr 2001:db8:dff0:6001::11 ip6 daddr 2001:db8:dff0:7002::4 tcp dport 443 accept
640     ip6 saddr 2001:db8:dff0:6001::11 ip6 daddr 2001:db8:dff0:7002::5 tcp dport 443 accept
641
642     # website: allow prometheus access
643     ip6 saddr 2001:db8:dff0:6001::13 ip6 daddr 2001:db8:dff0:7002::5 tcp dport 443 accept
644
645     # final reject
646     counter log prefix "final-fwd-services " reject
647 }
648
649 chain zone_services_external {
650     # basic network services (domain, ntp)
651     jump nis_clients
652
653     # any: authoritative nameserver for ACME DNS-01
654     ip6 daddr $ip6_ans meta l4proto {tcp, udp} th dport 53 accept
655
656     # fpx: outgoing http(s)
657     ip6 saddr 2001:db8:dff0:6001::3 tcp dport {80, 443} accept
658     ip saddr 192.0.2.3 tcp dport {80, 443} accept
659
660     # portal: allow api.peeringdb.com access
661     ip6 saddr 2001:db8:dff0:6001::2 ip6 daddr 2600:1f18:4110::/48 tcp dport 443 accept
662     # portal: allow whois (Doh! Some are still legacy only...)
663     ip6 saddr 2001:db8:dff0:6001::2 tcp dport 43 accept
664     ip saddr 192.0.2.2 tcp dport 43 accept
665     # portal: api.ixpdb.net 🤖
666     ip saddr 192.0.2.2 ip daddr 34.120.150.19 tcp dport 443 accept
667
668     # final reject
669     counter log prefix "final-fwd-services " reject
670 }

```

Peering LAN

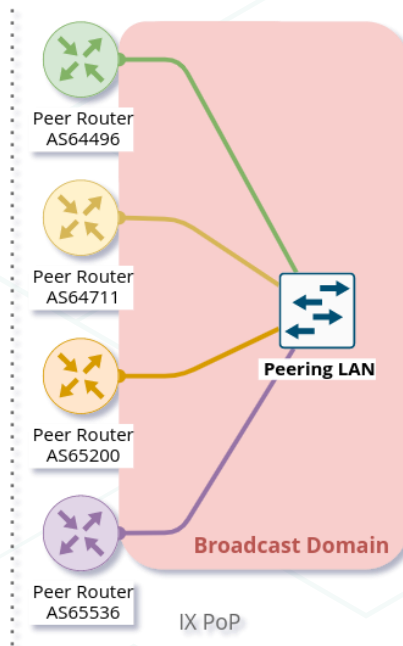
Challenges. Scalability.

- expand IX to connect additional switches and PoPs later without sacrificing fault tolerance and resource sharing
- plain Ethernet, link aggregation, and spanning tree might work for small LANs with few switches, but would become a major challenge in larger deployments since convergence in case of topology changes is slow and flexibility to steer packet forwarding is limited
- **DD-IX uses MP-BGP EVPN (Multiprotocol BGP Ethernet VPN) on top of VxLAN (Virtual Extensible LAN),**

Bundling links

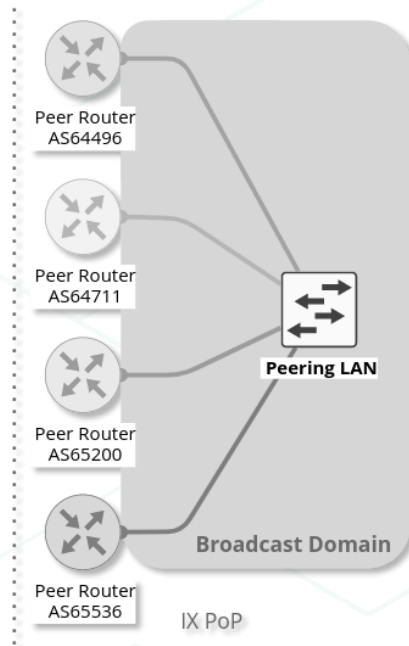
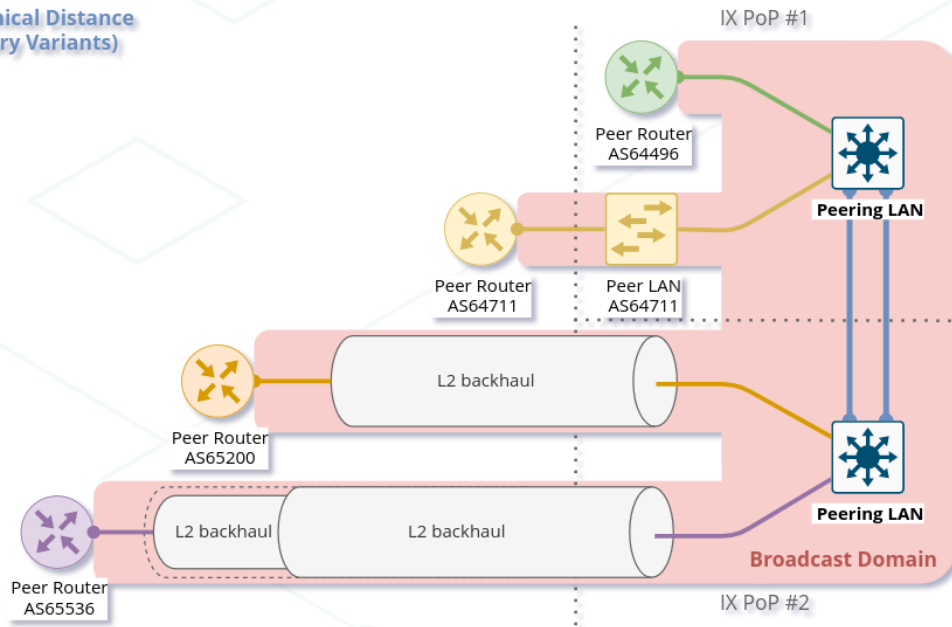
- Use of Equal-Cost Multi-Path (ECMP) instead of link aggregation
- Running a routing session on each individual link instead of aggregating links on layer 2 brings various advantages:
 - Links can be released from traffic by disabling the individual routing session. This allows link maintenances with zero packet loss since the routing protocol will gradually move the traffic instead of abruptly turn off ports.
 - Reconfigurations or even replacing the IGP of the underlay is possible without traffic disruption in the overlay.
 - BFD is performed independently on each link.

Protecting the peering LAN



Protecting the peering LAN

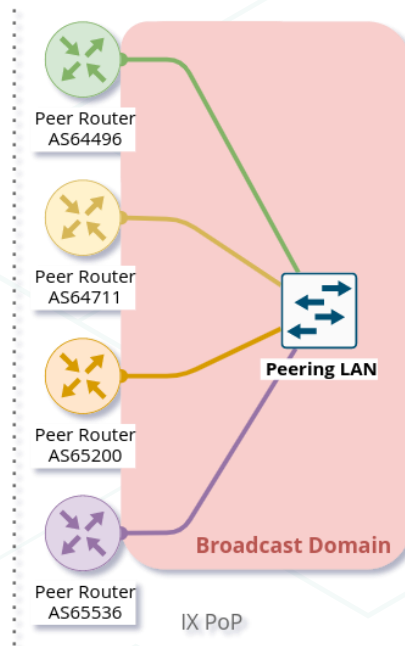
Geographical Distance
(Exemplary Variants)



Protecting the peering LAN

What could possibly go wrong?

1. The Peering LAN is just a bridge and all peering ports with L2 backhauls are building a large carrier-spanning (country-spanning? continent-spanning? ...) broadcast domain. No one has the intention to build a loop!
2. Every router in the Peering LAN can (unintentionally) disrupt the network by using incorrect layer 2 or layer 3 addresses, doing Proxy ARP, or doing other weird things.
3. Even if you trust your peers, there is still a risk that a router could be exploited by an attacker - it doesn't matter whether you trust your peers or not.



Example of incident @ amsix



Incident. Between 19:08 CET and 23:04 CET on November 22, 2023, and from 09:38 CET to 10:25 CET on November 23, 2023, customers connected to the Amsterdam infrastructure experienced active flapping of LACP and BGP sessions. **At the peak, the total traffic passing through the AMS-IX platform dropped to 2.1 Tb/s. BGP sessions were reduced from 885 to 550 (IPv4) and 800 to 450 (IPv6).**

What happened? LACP Leakage originated from a Juniper switch, which actively propagated LACP packets from customer-owned equipment. These LACP packets triggered the teardown of LACP LAGs for other customers. Resulting flapping induced resource starvation and full buffers, leading to RSVP timeout errors. Switches aggressively sent RSVP Path Error messages introducing further issues.

Why? Wrong ACLs.

Harden peering switch ports

Automatically assign the following configuration to all ports connecting peers having the status *Connected*

```
interface Ethernet11
  description PEER AS65372 peer1
  switchport access vlan 600
  ip access-group peer-eth11-ipv4 in
  ipv6 access-group peer-eth11-ipv6 in
  mac access-group peer-eth11 in
  no lldp transmit
  no lldp receive
  sflow enable
  storm-control broadcast level 0.01
  storm-control multicast level 0.01
  storm-control unknown-unicast level 0.01
```

Harden peering switch ports

```
mac access-list peer-eth11
  10 remark AS65372 allow arp broadcast
  11 permit vlan 600 0x000 00:53:42:8b:7b:20 00:00:00:00:00:00 ff:ff:ff:ff:ff:ff:
  20 remark AS65372 allow nd multicast
  21 permit vlan 600 0x000 00:53:42:8b:7b:20 00:00:00:00:00:00 33:33:ff:00:00:00:
  30 remark AS65372 drop other broadcasts or multicasts
  31 deny vlan 600 0x000 any 01:00:00:00:00:00 fe:ff:ff:ff:ff:ff log
  40 remark AS65372 allow lan access
  41 permit vlan 600 0x000 00:53:42:8b:7b:20 00:00:00:00:00:00 any arp
  42 permit vlan 600 0x000 00:53:42:8b:7b:20 00:00:00:00:00:00 any ip
  43 permit vlan 600 0x000 00:53:42:8b:7b:20 00:00:00:00:00:00 any ipv6
  100 remark AS65372 drop any
  101 deny any any log
! ...
mac address-table static 0053.428b.7b20 vlan 600 interface Ethernet11
```

L3 protection. Filtering IPv6 peering addresses is specifically challenging

(1) The IXP Manager does not allow to manage (static) link-local addresses.

(2) Autoconfigured link-local addresses require dedicated monitoring to discover them

```
ipv6 access-list peer-eth11-ipv6
 10 remark AS65372 IPv6 multicast (RFC 5963)
 11 permit ipv6 any host ff02::1
 12 permit ipv6 any host ff02::2
 13 permit ipv6 any host ff02::16
 14 permit ipv6 any ff02::1:ff00:0/104
 15 deny ipv6 any ff00::/8 log
 20 remark AS65372 allow router IPv6
 21 permit vlan 600 0x000 ipv6 host 2001:db8:79::3c0c:1 any
 30 remark AS65372 deny IPv6 abuse
 31 deny vlan 600 0x000 ipv6 2001:db8:79::/64 2001:db8:79::/64 log
 100 remark allow any other traffic
 101 permit ipv6 any any
! ...
ip access-list peer-eth11-ipv4
 10 remark AS65372 allow router IP
 11 permit vlan 600 0x000 ip host 198.51.100.71 any
 20 remark AS65372 deny IP abuse
 21 deny vlan 600 0x000 ip 198.51.100.64/26 198.51.100.64/26 log
 100 remark allow any other traffic
 101 permit ip any any
```

Operating systems and system isolation

Operating Systems on Servers

Alpine Linux for routers

- runs from tmpfs
- small overhead
- Linux routing: Kernel + BIRD



<https://github.com/dd-ix/ddix-ansible-ixp>
<https://github.com/dd-ix/ddix-ansible-rs>

NixOS for everything else

- declarative and reproducible
- rollbacks are easy
- fast if you know how



<https://github.com/dd-ix/nix-config>

Use Ansible to glue for cross-system automation



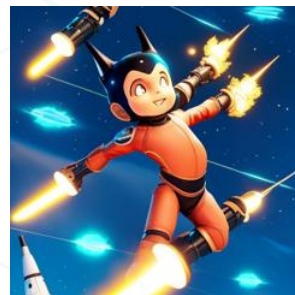
ANSIBLE



NixOS

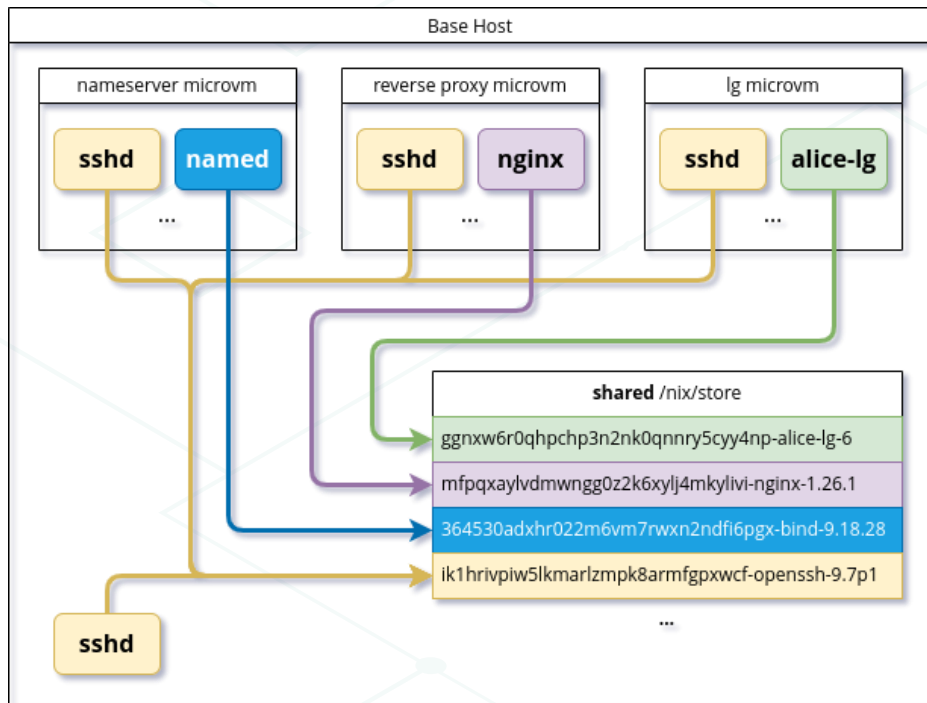
Service isolation

- currently, only single management host
- *MicroVM*'s for every service
 - KVM-based virtualization
 - diskless/tmpfs
 - mount points for persistent data
 - well integrated into NixOS thanks to @astro



github.com/astro/microvm.nix

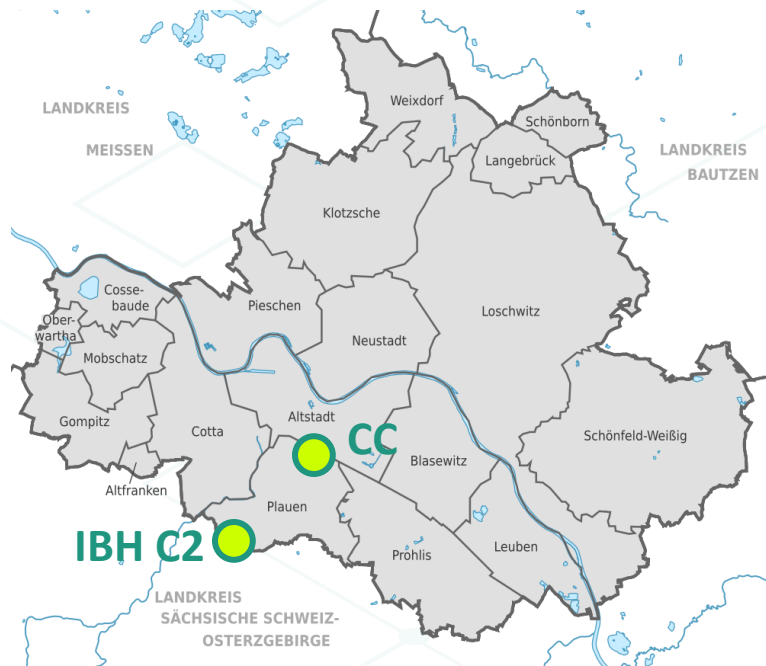
Service isolation



Mapping NixOS store read-only into the microvm containers using the virtiofs filesystem, results in almost no storage overhead compared to a single machine setup.

Points of Presences (PoPs)

DD-IX Locations



SACHSEN ENERGIE CENTER (SEC)

SachsenGigaBit GmbH

Friedrich-List-Platz 2

01069 Dresden

COSCHÜTZ 2 (C2)

IBH IT-Service GmbH

Heilbronner Str. 20

01189 Dresden

How does a switch get into a rack?



How does a switch get into a rack?



How does a switch get into a rack?



How does a switch get into a rack?



How does a switch get into a rack?



What happens if a peer connect?

- Establish a cross-connect via data center operator.
- Register MAC address of your device.
- Switch port changes state.
- Establish BGP session(s) to peer and route server. How hard can this be?

Mon, 21 Oct 2024 13:23:01

“Physikalische Verbindung steht, Interface in deine Richtung ist eingerichtet. MAC -Adresse meiner L3 Instanz ist: E4-81-84-C2-A3-34 Sobald das Interface hoch kommt, rede ich auch schon eBGP mit dir (ttl-security 1, multihop 255). Md5 key hab ich noch keinen eingetragen, kannst du mir dann ja aber geben, sobald alles geht.”

Mon, 21 Oct 2024 13:23:01

Mon, 21 Oct 2024 13:31:11

“wir sehen euch:

ixp-cc-sw01#show int eth13 transceiver dom

mA: milliamperes, dBm: decibels (milliwatts), C: Celsius, V:
Volts

Port 13

Temperature	41.78 C
-------------	---------

Voltage	3.26 V
---------	--------

Ethernet13 (Channel 1)

TX bias current	31.66 mA
-----------------	----------

Optical TX power	-0.52 dBm
------------------	-----------

Optical RX power	-2.88 dBm”
------------------	------------

Mon, 21 Oct 2024 13:23:01

Mon, 21 Oct 2024 13:31:11

“

> MAC -Adresse meiner L3 Instanz ist: XX
> Sobald das Interface hoch kommt, rede ich auch schon eBGP
> mit dir (ttl-security 1, multihop 255). Md5 key hab ich noch
> keinen eingetragen, kannst du mir dann ja aber geben, sobald
> alles geht.

<Password>

(du solltest das eigentlich im IXP Manager sehen)

Ich starte gleich unsere Automatisierung um euren Port ins produktive VLAN zu schieben.“

Mon, 21 Oct 2024 13:23:01

Mon, 21 Oct 2024 13:31:11

Mon, 21 Oct 2024 13:38:44

“entweder ich habe irgendwo was überlesen oder ich habe noch keinen Zugang zum IXP Manager.

Sollten wir schon einen haben, gib mir bitte die Mail, auf die der admin eingetragen ist.“

Mon, 21 Oct 2024 13:23:01

Mon, 21 Oct 2024 13:31:11

Mon, 21 Oct 2024 13:38:44

Mon, 21 Oct 2024 14:01:00

“euer Router ist per ICMP Echo erreichbar :-)

[IXP Manager]

Sorry, das hatten wir noch nicht besprochen bzw. war noch kein Nutzer eingerichtet *facepalm*. Ist jetzt auch angelegt.

”

Mon, 21 Oct 2024 13:23:01

Mon, 21 Oct 2024 13:31:11

Mon, 21 Oct 2024 13:38:44

Mon, 21 Oct 2024 14:01:00

Mon, 21 Oct 2024 14:11:33

“authentication key habe ich eingerichtet, IXP Manager mache ich gleich. BGP kommt nicht hoch. Bekomme einen *openfail*

Ich habe dir die falsche MAC gegeben, nimm bitte die hier: XX
”

Mon, 21 Oct 2024 13:23:01

Mon, 21 Oct 2024 13:31:11

Mon, 21 Oct 2024 13:38:44

Mon, 21 Oct 2024 14:01:00

Mon, 21 Oct 2024 14:11:33

Mon, 21 Oct 2024 14:27:49

“

> authentication key habe ich eingerichtet, IXP Manager mache
> ich gleich. BGP kommt nicht hoch. Bekomme einen
> *openfail*

das waren noch unsere L3 Filter (MAC Adresse... ;)

> Ich habe dir die falsche MAC gegeben, nimm bitte die hier: XX
Ah! Uch hatte mich schon gewundert weil ich diese initial in der
Quarantäne noch gesehen hatte.

Hast du in IPv6 das gleiche Secret hinterlegt? Ich sehe SYN requests
auf den RS in IPv6 (mit MD5) aber keinen vollständigen
Verbindungsaufbau.”

Mon, 21 Oct 2024 13:23:01

Mon, 21 Oct 2024 13:31:11

Mon, 21 Oct 2024 13:38:44

Mon, 21 Oct 2024 14:01:00

Mon, 21 Oct 2024 14:11:33

Mon, 21 Oct 2024 14:27:49

Mon, 21 Oct 2024 14:47:31

“Ja, ich nutze für IPv6 das gleiche Secret. Grad nochmal geschaut.

Habe jetzt am rs01 das Secret rausgenommen um zu testen, ob es überhaupt daran liegt oder vielleicht an einem ACL Filter, o.ä.

Und ich bin heute ein bisschen durch den Wind. Die Email Adresse lautet natürlich XX.”

Mon, 21 Oct 2024 13:23:01

Mon, 21 Oct 2024 13:31:11

Mon, 21 Oct 2024 13:38:44

Mon, 21 Oct 2024 14:01:00

Mon, 21 Oct 2024 14:11:33

Mon, 21 Oct 2024 14:27:49

Mon, 21 Oct 2024 14:47:31

Mon, 21 Oct 2024 14:53:41

„wenn ich tcpdump den MD5 Secret mit übergebe sehe ich das euer Router

und unserer RS01 korrekte MD5 Hash bilden:

<tcpdump>

Ich sah dein SYN, wir schickten ein SYN+ACK zurück aber dann kommt nix mehr. Hast du hier eine ACL auf deinem Router?

Aktuell kommt es (wie du geschrieben hast) ohne Secret und wird deswegen auf dem RS01 gedropped.,,

Mon, 21 Oct 2024 13:23:01

Mon, 21 Oct 2024 13:31:11

Mon, 21 Oct 2024 13:38:44

Mon, 21 Oct 2024 14:01:00

Mon, 21 Oct 2024 14:11:33

Mon, 21 Oct 2024 14:27:49

Mon, 21 Oct 2024 14:47:31

Mon, 21 Oct 2024 14:53:41

Mon, 21 Oct 2024 15:13:46

„als ich den Fehler gesehen habe war die erste Aktion, die ACL von meinem Interface runter zu nehmen und zu schauen, ob es daran liegt. Lag es nicht...

Ich werde mir das Morgen früh nochmal zu Gemüte führen, muss jetzt leider Feierabend machen.

Ich stelle solange die ipv6 Peering sessions auf Down. IPv4 reden wir ja erfolgreich.

Ich hoffe, das passt für dich.“

Mon, 21 Oct 2024 13:23:01

Mon, 21 Oct 2024 13:31:11

Mon, 21 Oct 2024 13:38:44

Mon, 21 Oct 2024 14:01:00

Mon, 21 Oct 2024 14:11:33

Mon, 21 Oct 2024 14:27:49

Mon, 21 Oct 2024 14:47:31

Mon, 21 Oct 2024 14:53:41

Mon, 21 Oct 2024 15:13:46

Mon, 22 Oct 2024 08:09:28

„wie versprochen habe ich mich heute früh nochmal hingesezt und mir das Problem angeschaut.

Wir hatten ein kleines sync Problem bei der Anpassung unserer CPM Filter. Das ist gelöst und die IPv6 Peers laufen jetzt auch.

Damit ist von unserer Seite der IXPeer vollständig eingerichtet.

Vielen Dank für deine Unterstützung (und Geduld).“

Interested into more details?

IXP-from-Scratch: Building a New IX



Thomas Liske — 12 Aug 2024

Contributors: [Marcel Koch](#), [Tassilo Tanneberger](#), [Matthias Wählich](#)
5 min read

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In this new series "IXP from Scratch", we report about our endeavor of building DD-IX, an Internet Exchange located in the city of Dresden, Germany. DD-IX is driven by a grassroots community that loves contributing to a resilient and efficient Internet. We start with some basic background and will share technical design decisions in upcoming articles.

Believe it or not, the Internet is still a community project

The Internet is a network of networks and Internet Exchange Points (IXP) are a crucial part of the underlying infrastructure. They facilitate public and local interconnections between networks, which increases resiliency because of higher interconnectivity and reduces delays because of vicinity. They are a place where networks meet technically and non-technically.

More from this author

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IXP-from-Scratch: The Peering LAN

Thomas Liske
30 Sep 2024



IXP-from-Scratch: Network and Security Design

Thomas Liske
26 Aug 2024

Want to get involved?

Talk to us! Our chair is heavily involved at DD-IX.

Meet us! DD-IX weekly tech meeting, every Tuesday, 5:30pm at APB 3080.

Write us! contact@dd-ix.net

